

A Counterexample for Zero APEPs in Infinite ℓ_1 -Sums

Automated Math Discovery Smoke Test

June 8, 2026

Source and Open Question

This packet concerns Question 8.1 from:

Ramon J. Aliaga, Luis C. Garcia-Lirola, Juan Guerrero-Viu, Matias Raja, Abraham Rueda Zoca, *Almost preserved extreme points*, arXiv:2510.21234.

The source location is page 37 of the paper.

ALMOST PRESERVED EXTREME POINTS

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8. CONCLUDING REMARKS AND OPEN QUESTIONS

In this section we collect some comments and open questions which are derived from our work.

From our study of APEP in the unit ball of ℓ_1 -sums of spaces, the following remains open.

Question 8.1. Let $\{X_i : i \in I\}$ an arbitrary infinite family of Banach spaces and let X be its ℓ_1 -sum. If $0 \in B_X$ is APEP in B_X , must there exist i such that $0 \in X_i$ is APEP in B_{X_i} ?

For the case of ℓ_p -sums, in Proposition 4.9 we have characterised when the norm-one elements are APEP of the unit ball. However, we were unable to describe those APEP whose norm is strictly smaller than 1.

Question 8.1 asks:

Let $\{X_i : i \in I\}$ be an arbitrary infinite family of Banach spaces and let X be its ℓ_1 -sum. If $0 \in B_X$ is APEP in B_X , must there exist i such that $0 \in X_i$ is APEP in B_{X_i} ?

We give a negative answer.

Definitions

Let C be a bounded closed convex subset of a Banach space E . A point $x \in C$ is an almost preserved extreme point, abbreviated APEP, if every relatively weakly open subset W of C containing x contains a non-empty slice of C . For the unit ball, a slice has the form

$$S(B_E, f, \alpha) = \{y \in B_E : f(y) > \sup f(B_E) - \alpha\},$$

where $f \in E^*$ and $\alpha > 0$.

Counterexample

Theorem 1. *There is a countably infinite family of Banach spaces (X_n) such that 0 is an APEP of the unit ball of $X = (\bigoplus_{n=1}^{\infty} X_n)_1$, but 0 is not an APEP of the unit ball of any coordinate X_n .*

Proof. For each $n \geq 1$, let $X_n = H_n = \mathbb{R}^n$ with its Euclidean norm, and put

$$X = \left(\bigoplus_{n=1}^{\infty} H_n \right)_1.$$

Then

$$X^* = \left(\bigoplus_{n=1}^{\infty} H_n \right)_{\infty},$$

where we identify each H_n^* with H_n by the inner product.

First, 0 is not an APEP of B_{H_n} for any fixed n . Indeed, in finite dimension the weak and norm topologies coincide, so

$$W_n = \{h \in B_{H_n} : \|h\| < 1/2\}$$

is a relatively weakly open neighbourhood of 0 in B_{H_n} . No non-empty slice of B_{H_n} is contained in W_n : if $S(B_{H_n}, f, \alpha)$ is non-empty, then either it is all of B_{H_n} in the degenerate case, or it contains a unit vector at which f attains its norm. In both cases it contains a point of norm 1.

It remains to show that 0 is an APEP of B_X . Let W be a relatively weakly open neighbourhood of 0 in B_X . After shrinking W , we may assume that

$$W = \{x \in B_X : |x_j^*(x)| < \varepsilon, \quad j = 1, \dots, m\}$$

for some $x_1^*, \dots, x_m^* \in X^*$ and $\varepsilon > 0$. Write

$$x_j^* = (a_{j,n})_{n \geq 1}, \quad a_{j,n} \in H_n,$$

and let

$$M = \max_{1 \leq j \leq m} \|x_j^*\|.$$

If $M = 0$, any non-empty slice of B_X is contained in W , so assume $M > 0$.

Choose $n > m$. Since $\dim H_n = n$, there is a unit vector $u \in H_n$ orthogonal to the finite set

$$a_{1,n}, a_{2,n}, \dots, a_{m,n}.$$

Define $\varphi \in X^*$ by

$$\varphi((h_k)_{k \geq 1}) = \langle h_n, u \rangle.$$

Choose $\delta > 0$ so small that

$$M(\sqrt{2\delta} + \delta) < \varepsilon.$$

Consider the slice

$$S = \{x = (h_k) \in B_X : \varphi(x) > 1 - \delta\}.$$

This slice is non-empty, since it contains the vector supported only at coordinate n with value u .

We claim that $S \subseteq W$. Let $x = (h_k) \in S$. Since

$$1 - \delta < \langle h_n, u \rangle \leq \|h_n\| \leq \sum_{k=1}^{\infty} \|h_k\| \leq 1,$$

we have

$$\sum_{k \neq n} \|h_k\| < \delta.$$

Also, using $\|h_n\| \leq 1$ and $\langle h_n, u \rangle > 1 - \delta$,

$$\|h_n - u\|^2 = \|h_n\|^2 + 1 - 2\langle h_n, u \rangle \leq 2\delta.$$

For each $j = 1, \dots, m$, orthogonality gives

$$\langle a_{j,n}, h_n \rangle = \langle a_{j,n}, h_n - u \rangle.$$

Therefore

$$\begin{aligned} |x_j^*(x)| &\leq |\langle a_{j,n}, h_n \rangle| + \sum_{k \neq n} |\langle a_{j,k}, h_k \rangle| \\ &\leq M \|h_n - u\| + M \sum_{k \neq n} \|h_k\| \\ &< M(\sqrt{2\delta} + \delta) < \varepsilon. \end{aligned}$$

Thus $x \in W$ for every $x \in S$, so $S \subseteq W$.

Every relatively weakly open neighbourhood of 0 in B_X contains a non-empty slice of B_X . Hence 0 is an APEP of B_X , even though no coordinate has 0 as an APEP. This gives a negative answer to Question 8.1. $\square$

Verification Notes

The argument is purely analytic. The only quantitative estimate is the Hilbert-space fact that if $\langle h, u \rangle > 1 - \delta$ with $\|h\| \leq 1 = \|u\|$, then $\|h - u\| \leq \sqrt{2\delta}$. The script `code/check_slice_estimate.py` performs randomized finite dimensional checks of the orthogonal-vector construction and this bound. It is a sanity check only; the proof above is the certificate.

Limitations and Novelty Check

This packet answers the source question as stated and in its intended infinite-sum setting: the counterexample is a genuine infinite ℓ_1 -sum, not a degenerate one-dimensional edge case. I have not performed a literature search beyond the source paper and the local pipeline corpus, so the novelty status is “candidate” pending human and literature review.